Notes: Pavcnik (RES, 2002)

Trade Liberalization, Exit, and Productivity Improvements: Evidence from Chilean Plants

Contents

1 Big picture

- **Question.** Does trade liberalization raise plant productivity, and if so, does the gain come from continuing plants becoming more productive, from low-productivity plants exiting, or from output and resources being reallocated toward more productive producers?
- **Setting.** The paper studies Chilean manufacturing plants from 1979 to 1986, after Chile's large trade liberalization of the late 1970s. The setting is useful because Chile sharply reduced tariffs and eliminated most non-tariff barriers before the sample period, and plants were still adjusting during the early 1980s.
- **Core empirical challenge.** Productivity is not directly observed. It must be recovered from a production function, but input choices and plant exit are endogenous to productivity. More productive plants choose inputs differently and are less likely to exit.
- **Methodological solution.** The paper estimates production functions using a semiparametric Olley–Pakes-style approach that controls for simultaneity through investment and controls for selection through the probability of plant survival.
- **Identification of trade effects.** The paper compares productivity changes across import-competing, export-oriented, and nontraded sectors over time. The nontraded sector acts as the main comparison group for shocks not directly related to foreign competition.
- **Main result: within-plant productivity.** Plants in import-competing sectors experience productivity growth about 3% to 10% higher than plants in nontraded sectors. This is interpreted as evidence that foreign competition forced previously protected plants to "trim their fat."
- **Main result: exit.** Exiting plants are about 8% less productive than surviving plants. Exit therefore contributes to aggregate productivity growth by removing inefficient producers.
- **Main result: reallocation.** Aggregate productivity growth also comes from the reallocation of market shares and resources from less productive to more productive plants. This reallocation channel is especially important for export-oriented and import-competing sectors.
- **Economic takeaway.** The gains from trade liberalization are not only scale-economy gains. In a heterogeneous-plant economy, trade can raise productivity through within-plant discipline, selective exit, and reallocation toward efficient producers.

2 Incremental contribution of the paper

- **From trade liberalization to plant dynamics.** Earlier empirical work often compared productivity before and after trade reform. This paper emphasizes that aggregate productivity gains may come from heterogeneous plant responses: some plants improve, some exit, and output shifts toward more efficient plants.
- **A cleaner trade-identification strategy.** Instead of relying only on time-series before-after comparisons, the paper uses variation over time and across trade orientation. Import-competing and export-oriented sectors are compared with nontraded sectors, helping separate trade-related productivity changes from macro shocks.

- **A production-function measurement contribution.** The paper imports the Olley and Pakes (1996) semiparametric production-function method into the trade-productivity literature. This improves on OLS and fixed effects approaches that can produce biased input coefficients.
- **Explicit correction for plant exit.** A key innovation is to correct for selection bias from plant liquidation. Because less productive plants are more likely to exit, ignoring exit biases production-function estimates and therefore biases measured productivity.
- **Evidence on both within-plant and between-plant channels.** The paper separates within-plant productivity improvements from aggregate gains due to exit and market-share reallocation. This distinction anticipates the modern heterogeneous-firm view of trade gains.
- **A developing-country plant-level application.** The Chilean plant census includes relatively small plants, allowing the paper to study adjustment margins that many firm-level datasets miss.
- **Policy contribution.** The paper shows that productivity gains from trade liberalization can be real but costly: they operate partly through plant exit, labour displacement, and resource reallocation. Thus, barriers to exit and reallocation are central to the success of trade reform.

3 Data and measurement

3.1 Chilean trade liberalization and sample period

Chile undertook a large trade liberalization between 1974 and 1979. Tariffs that often exceeded 100% in 1974 were reduced to a uniform 10% ad valorem tariff by 1979. Chile remained broadly committed to free trade during the 1980s, though tariffs temporarily increased during the 1982–1983 recession before declining again.

The plant-level panel covers 1979–1986. This period does not include the initial tariff cuts, but it captures the adjustment phase after plants had been exposed to foreign competition and after the reform became more credible.

Interpretation: the design studies medium-run adjustment after reform rather than the immediate announcement effect of reform.

3.2 Plant-level manufacturing census

The data come from a census of Chilean manufacturing plants with at least ten employees. The unit of observation is a plant, although more than 90% of plants are single-plant establishments. The final sample contains 4,379 plants and 25,491 plant-year observations.

Key variables include output, value added, capital, investment, materials, skilled labour, and unskilled labour. Monetary variables are expressed in constant 1980 pesos.

Interpretation: the plant-level nature of the data is essential because trade liberalization affects plants differently even within the same industry.

3.3 Trade orientation

Each plant is assigned to a four-digit ISIC industry. Industries are classified as:

- **export-oriented** if exports exceed 15% of total output;
- **import-competing** if imports exceed 15% of total domestic output;
- **nontraded** otherwise.

The classification is based on Chilean trade balances. The paper checks robustness using other measures of exposure to trade, including import-output ratios, tariffs, and exchange rates.

Interpretation: import-competing plants are the main treated group for foreign-competition discipline. Nontraded plants provide a comparison group for macro shocks that affect all sectors.

3.4 Productivity measurement

The paper estimates plant productivity from a Cobb–Douglas gross-output production function:

$$y_{it} = \beta_0 + \beta_x x_{it} + \beta_k k_{it} + e_{it}, \qquad e_{it} = \omega_{it} + \mu_{it}.$$

Here y_{it} is log output, x_{it} contains variable inputs such as skilled labour, unskilled labour, and materials, and k_{it} is capital. The term ω_{it} is plant productivity known to the plant but unobserved by the econometrician; μ_{it} is an unexpected productivity shock.

The measured productivity index is the difference between actual and predicted output, normalized relative to a reference plant in the base year 1979:

$$pr_{it} = y_{it} - \hat{\beta}_{sl} l^s_{it} - \hat{\beta}_{ul} l^u_{it} - \hat{\beta}_m m_{it} - \hat{\beta}_k k_{it} - (y_r - \hat{y}_r).$$

Interpretation: a plant has high measured productivity if it produces more output than predicted given its measured inputs.

4 Models and empirical specifications

4.1 Why OLS and fixed effects are problematic

OLS is biased because more productive plants choose inputs differently. If high-productivity plants hire more workers, use more materials, or invest more, input choices are correlated with unobserved productivity. This creates simultaneity bias.

Fixed effects remove only time-invariant productivity differences. This is problematic in a trade-liberalization setting because the object of interest is precisely how plant productivity changes over time.

Exit creates an additional selection problem. Plants continue producing only if expected future profits exceed liquidation value. Less productive plants are more likely to exit. If survival is related to both productivity and capital, ignoring exit biases the capital coefficient and therefore measured productivity.

4.2 Olley–Pakes semiparametric structure

The paper follows the dynamic plant framework of Ericson and Pakes (1995) and Olley and Pakes (1996). A plant first decides whether to exit or continue. Conditional on continuing, it chooses inputs and investment.

The exit rule is:

$$X_t = \begin{cases} 1, & \omega_t \geq \bar{\omega}_t(k_t), \\ 0, & \text{otherwise,} \end{cases}$$

where $X_t = 1$ means the plant survives.

The investment rule is:

$$i_t = i_t(\omega_t, k_t).$$

If investment is monotone in productivity, the rule can be inverted:

$$\omega_t = h_t(i_t, k_t).$$

This allows the econometrician to control for unobserved productivity using observable investment and capital.

4.3 First stage: coefficients on variable inputs

Substituting the inverted investment rule into the production function yields:

$$y_t = \beta_x x_t + \lambda_t(k_t, i_t) + \mu_t,$$

where

$$\lambda_t(k_t, i_t) = \beta_0 + \beta_k k_t + h_t(k_t, i_t).$$

The function $\lambda_t(\cdot)$ is approximated by a polynomial in capital and investment. This controls for plant productivity and produces consistent estimates of the coefficients on variable inputs.

Interpretation: investment serves as a proxy for the plant's private productivity information.

4.4 Survival correction and capital coefficient

The capital coefficient requires an additional correction because capital is related to expected future productivity and because the sample only contains surviving plants. The probability that a plant stays in the market is modeled as:

$$P_t = \Pr(X_{t+1} = 1) = P_t(k_t, i_t).$$

The final stage estimates a production function that controls for expected productivity and survival probability:

$$y_{t+1} - \hat{\beta}_x x_{t+1} = \beta_0 + \beta_k k_{t+1} + \Phi(\hat{\lambda}_t - \beta_k k_t, \hat{P}_t) + \xi_{t+1} + \mu_{t+1}.$$

Interpretation: the survival probability corrects for the fact that observed plants are selected survivors rather than a random sample of all plants.

4.5 Aggregate productivity decomposition

The aggregate industry productivity index is a market-share-weighted average of plant productivity:

$$W_t = \sum_i s_{it} pr_{it},$$

where s_{it} is plant i's output share in its industry.

Following Olley and Pakes, it can be decomposed as:

$$W_t = \bar{pr}_t + \sum_i (s_{it} - \bar{s}_t)(pr_{it} - \bar{pr}_t).$$

The first term is unweighted average productivity. The second term is the covariance between output share and productivity.

Interpretation: a rising covariance means that more output is produced by more productive plants. This is the reallocation channel.

4.6 Difference-in-differences productivity regressions

The paper studies plant-level productivity using:

$$pr_{it} = \alpha_0 + \alpha_1 Time_t + \alpha_2 Trade_i + \alpha_3 (Trade_i \times Time_t) + \alpha_4 Z_{it} + u_{it}.$$

Here $Trade_i$ contains indicators for export-oriented and import-competing sectors, while nontraded sectors are the excluded group. $Time_t$ contains year indicators. Z_{it} includes plant characteristics such as industry affiliation and exit indicators.

The coefficients on the interactions $Trade_i \times Time_t$ measure productivity changes in traded sectors relative to nontraded sectors in the same year.

Interpretation: the import-competing interaction coefficients are the core estimates of productivity effects attributable to exposure to import competition.

5 Identification and empirical design

5.1 What the design can identify

The design identifies whether productivity evolved differently in sectors directly exposed to trade relative to nontraded sectors. The most important comparison is import-competing versus non-traded plants after liberalization.

A positive coefficient for import-competing plants means that plants in previously shielded sectors became more productive relative to plants in nontraded sectors. Because the paper controls for year effects, common macro shocks are absorbed.

Interpretation: the evidence is strongest for relative productivity changes associated with import competition, not for the full general-equilibrium welfare effect of trade liberalization.

5.2 Main identifying assumptions

The design requires that, absent trade exposure, productivity in import-competing and nontraded sectors would not have diverged systematically for other reasons. This is a parallel-trends-style assumption, although the paper lacks plant-level pre-1979 panel data to test it directly.

The design also assumes that macro shocks, especially the 1982–1983 recession, do not affect import-competing and nontraded sectors in a way that exactly mimics trade liberalization effects. Year fixed effects capture common shocks, but sector-specific recession exposure remains a concern.

Interpretation: the paper strengthens credibility by using several robustness checks, but the setting is observational rather than experimental.

5.3 Exchange-rate and demand concerns

A real exchange-rate appreciation could reduce measured productivity in traded sectors by reducing demand and capacity utilization, while raising demand for nontraded goods. This could contaminate the trade-orientation coefficients.

The paper addresses this in three ways:

- showing that the correlation between productivity growth and output growth is small;
- showing that inventories do not fluctuate in a pattern consistent with the spare-capacity story;
- estimating regressions that relate productivity to the real exchange rate and its interaction with trade orientation.

The evidence suggests that exchange-rate movements may help explain the temporary productivity decline in export-oriented sectors in 1980–1981, but they do not explain the persistent import-competing productivity gains.

5.4 Interpreting measured productivity

Measured productivity is based on real output values deflated by industry-level price indexes, not plant-level physical output. If plants charge different markups, measured productivity may combine true technical efficiency and price/markup variation.

The paper notes this caveat and uses plant fixed effects in some regressions to control for time-invariant plant-specific markups.

Interpretation: the results are best read as improvements in revenue productivity or measured productivity, with the usual caution that price and markup differences may matter.

6 Tables and figures

6.1 Table 1: Plants active in 1979 but not in 1986

Read:

- Table 1 documents the scale and composition of plant exit between 1979 and 1986.
- Overall, 35.2% of plants active in 1979 were no longer active by 1986.
- Exiting plants accounted for 25.2% of 1979 labour, 7.8% of capital, 13.5% of investment, 15.5% of value added, and 15.6% of output.
- Among exiting plants, 40.1% were import-competing and 47.0% were nontraded; only 12.9% were export-oriented.
- Within sectors, 41.6% of export-oriented plants, 38.3% of import-competing plants, and 31.6% of nontraded plants exited.

Economic message: Exit is not a marginal phenomenon. Plant liquidation is central to the post-liberalization adjustment process, so production-function estimation and aggregate productivity analysis must explicitly account for selection and reallocation.

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TABLE 1

Plants active in 1979 but not in 1986

Trade orientation	Share of plants	Share of labour	Share of capital	Share of investment	Share of value added	Share of output
Exiting plants of a given trade orientation as a share of all plants active in 1979						
All trade orientations	0·352	0·252	0·078	0·135	0·155	0·156
Export-oriented	0·045	0·049	0·009	0·039	0·023	0·023
Import-competing	0·141	0·108	0·029	0·047	0·068	0·065
Nontraded	0·165	0·095	0·040	0·049	0·064	0·067
Exiting plants of a given trade orientation as a share of all exiting plants						
Export-oriented	0·129	0·194	0·117	0·289	0·149	0·148
Import-competing	0·401	0·429	0·369	0·350	0·436	0·419
Nontraded	0·470	0·377	0·513	0·361	0·415	0·432
Exiting plants of a given trade orientation as a share of all plants active in 1979 in the corresponding trade sector						
Export-oriented	0·416	0·298	0·030	0·172	0·121	0·128
Import-competing	0·383	0·263	0·093	0·149	0·183	0·211
Nontraded	0·316	0·224	0·104	0·107	0·147	0·132

Note: This figure also includes plants that exited after the end of 1979, but before the end of 1980 and were excluded in the estimation because of missing capital variable.

Table 1: Plants active in 1979 but not in 1986.

TABLE 2

Estimates of production functions

		Balanced panel				Full sample				
		OLS		Fixed effects		OLS		Fixed effects		
		(1)		(2)		(3)		(4)		
		Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	Coe
Food processing	Unskilled labour	0·152	0·007	0·185	0·012	0·178	0·006	0·210	0·010	0·15
	Skilled labour	0·127	0·006	0·027	0·008	0·131	0·006	0·029	0·007	0·09
	Materials	0·790	0·004	0·668	0·008	0·763	0·004	0·646	0·007	0·73
	Capital	0·046	0·003	0·011	0·007	0·052	0·003	0·014	0·006	0·07
	N	6432				8464				708
Textiles	Unskilled labour	0·187	0·011	0·240	0·017	0·229	0·009	0·245	0·015	0·21
	Skilled labour	0·184	0·010	0·088	0·014	0·183	0·009	0·088	0·012	0·17
	Materials	0·667	0·007	0·564	0·011	0·638	0·006	0·558	0·009	0·63
	Capital	0·056	0·005	0·015	0·012	0·059	0·004	0·019	0·011	0·05
	N	3689				5191				426
Wood	Unskilled labour	0·233	0·016	0·268	0·026	0·247	0·013	0·273	0·022	0·19
	Skilled labour	0·121	0·015	0·040	0·021	0·146	0·012	0·047	0·018	0·13
	Materials	0·685	0·010	0·522	0·014	0·689	0·008	0·554	0·011	0·67
	Capital	0·055	0·007	0·023	0·018	0·050	0·006	−0·002	0·016	0·10
	N	1649				2705				215
Paper	Unskilled labour	0·218	0·024	0·258	0·033	0·246	0·021	0·262	0·029	0·19
	Skilled labour	0·190	0·018	0·022	0·027	0·180	0·016	0·050	0·023	0·20
	Materials	0·624	0·013	0·515	0·025	0·597	0·011	0·514	0·021	0·60
	Capital	0·074	0·010	0·031	0·025	0·085	0·009	0·031	0·023	0·06
	N	1039				1398				114
Chemicals	Unskilled labour	0·033	0·014	0·239	0·022	0·067	0·013	0·246	0·020	0·03
	Skilled labour	0·211	0·013	0·079	0·018	0·213	0·012	0·090	0·017	0·19
	Materials	0·691	0·009	0·483	0·013	0·698	0·008	0·473	0·013	0·67
	Capital	0·108	0·008	0·032	0·014	0·089	0·007	0·036	0·013	0·12
	N	2145				2540				208
Glass	Unskilled labour	0·353	0·032	0·405	0·045	0·406	0·030	0·435	0·043	0·42
	Skilled labour	0·285	0·035	0·068	0·042	0·226	0·031	0·056	0·038	0·18
	Materials	0·523	0·022	0·360	0·026	0·544	0·019	0·403	0·024	0·52
	Capital	0·092	0·041	−0·015	0·036	0·093	0·011	−0·013	0·030	0·14
	N	623				816				66
Basic metals	Unskilled labour	0·080	0·037	0·137	0·070	0·105	0·037	0·174	0·072	0·12
	Skilled labour	0·158	0·034	0·008	0·070	0·156	0·034	0·006	0·072	0·11
	Materials	0·789	0·017	0·572	0·040	0·771	0·016	0·567	0·039	0·72
	Capital	0·030	0·014	0·033	0·030	0·025	0·013	0·034	0·032	0·11
	N	306				362				25
Machinery	Unskilled labour	0·186	0·013	0·225	0·018	0·199	0·012	0·238	0·016	0·17
	Skilled labour	0·238	0·011	0·130	0·016	0·222	0·010	0·112	0·014	0·20
	Materials	0·611	0·008	0·530	0·012	0·619	0·007	0·548	0·010	0·61
	Capital	0·078	0·006	0·057	0·013	0·078	0·005	0·047	0·013	0·05
	N	3025				4015				326

Table 2: Estimates of production functions.

6.2 Table 2: Estimates of production functions

Read:

- Table 2 compares OLS, fixed effects, and semiparametric production-function estimates across industries.

- OLS and fixed effects produce different input coefficients because they handle unobserved productivity very differently.
- The semiparametric estimates are designed to correct for simultaneity and selection bias.
- The capital coefficient often rises substantially under the semiparametric method. For example, in food processing, the capital coefficient is 0.079 under the semiparametric series method versus 0.052 under full-sample OLS and 0.014 under full-sample fixed effects.
- These differences matter because the productivity index is calculated using estimated input coefficients.

Economic message: Productivity measurement is not mechanical. Correcting for input endogeneity and plant survival changes the production-function estimates, especially the capital coefficient, and therefore changes measured productivity.

6.3 Table 2(a): Positive-investment robustness

Read:

- Olley–Pakes requires investment to be strictly positive for the monotonicity inversion to be exact.
- Table 2(a) compares estimates using all observations with estimates using only observations with strictly positive investment.
- Most coefficients are reasonably similar across the two samples.
- The paper uses this comparison to argue that including zero-investment observations is not driving the main productivity-trade results.

Economic message: The core conclusions are not simply an artifact of the technical investment-positivity requirement in the semiparametric estimator.

6.4 Table 3: Decomposition of aggregate productivity growth

Read:

- Table 3 decomposes aggregate productivity growth into unweighted average productivity and the covariance between output share and productivity.
- Manufacturing-wide aggregate productivity rises by 19.3% from 1979 to 1986.
- Of this, 6.6 percentage points come from unweighted productivity growth and 12.7 percentage points come from the covariance/reallocation component.
- Import-competing sectors show the largest 1979–1986 aggregate productivity gain, 31.9%, with a large covariance component of 21.3 percentage points.
- Nontraded sectors show much smaller gains: aggregate productivity rises only 6.2% by 1986.

Economic message: A large share of aggregate productivity growth comes from reallocating output toward more productive plants. The trade-productivity effect is therefore not only a within-plant story.

TABLE 2(a)

Comparison of the semiparametric estimates of production funct

		(1)		(2)
		Coef.	S.E.	Coef.
Food processing	Unskilled labour	0·153	0·007	0·081
	Skilled labour	0·098	0·009	0·119
	Materials	0·735	0·008	0·723
	Capital	0·079	0·034	0·070
	N	7085		2806
Textiles	Unskilled labour	0·215	0·012	0·183
	Skilled labour	0·177	0·011	0·166
	Materials	0·637	0·097	0·626
	Capital	0·052	0·034	0·056
	N	4265		1591
Wood	Unskilled labour	0·195	0·015	0·149
	Skilled labour	0·130	0·014	0·134
	Materials	0·679	0·010	0·654
	Capital	0·101	0·051	0·107
	N	2154		692
Paper	Unskilled labour	0·193	0·024	0·120
	Skilled labour	0·203	0·018	0·224
	Materials	0·601	0·014	0·594
	Capital	0·068	0·018	0·138
	N	1145		494
Chemicals	Unskilled labour	0·031	0·014	0·018
	Skilled labour	0·194	0·016	0·188
	Materials	0·673	0·012	0·666
	Capital	0·129	0·052	0·138
	N	2087		1247
Glass	Unskilled labour	0·426	0·035	0·400
	Skilled labour	0·183	0·036	0·132
	Materials	0·522	0·024	0·489
	Capital	0·142	0·053	0·113
	N	666		294
Basic metals	Unskilled labour	0·121	0·041	0·117
	Skilled labour	0·117	0·043	0·116
	Materials	0·727	0·032	0·753
	Capital	0·110	0·051	0·079
	N	255		158
Machinery	Unskilled labour	0·178	0·015	0·089
	Skilled labour	0·202	0·012	0·231
	Materials	0·617	0·009	0·626
	Capital	0·051	0·013	0·119
	[illegible]	[illegible]		[illegible]

Table 2(a): Comparison of semiparametric estimates.

TABLE 3

Decomposition of aggregate productivity growth

Industry	Year	Aggregate Productivity	Unweighted Productivity	Covariance	Industry	Year	Aggregate Productivity	Unweighted Productivity	C[illegible]
Food	79	0·000	0·000	0·000	Chemicals	79	0·000	0·000	
	80	0·005	0·008	−0·003		80	0·014	0·046	[illegible]
	81	0·008	0·058	−0·049		81	0·126	0·076	
	82	0·209	0·099	0·110		82	0·312	0·039	
	83	0·144	0·049	0·095		83	0·238	−0·050	
	84	0·116	0·044	0·072		84	0·156	−0·040	
	85	0·092	0·014	0·078		85	0·229	−0·033	
	86	0·179	0·129	0·050		86	0·432	−0·056	
Textiles	79	0·000	0·000	0·000	Glass	79	0·000	0·000	
	80	0·064	0·063	0·001		80	0·137	−0·036	
	81	0·148	0·119	0·029		81	0·109	−0·073	
	82	0·147	0·090	0·057		82	0·155	−0·044	
	83	0·075	0·063	0·012		83	0·231	−0·052	
	84	0·130	0·082	0·048		84	0·257	−0·071	
	85	0·136	0·095	0·041		85	0·193	−0·095	
	86	0·184	0·171	0·013		86	0·329	−0·011	
Wood	79	0·000	0·000	0·000	Basic metals	79	0·000	0·000	
	80	−0·052	−0·030	−0·022		80	−0·136	−0·022	[illegible]
	81	−0·125	−0·071	−0·054		81	−0·002	0·050	[illegible]
	82	0·070	−0·076	0·145		82	0·711	0·215	
	83	0·148	−0·051	0·198		83	0·343	0·030	
	84	0·169	0·038	0·131		84	0·153	−0·037	
	85	0·019	−0·038	0·058		85	0·228	−0·153	
	86	−0·035	0·045	−0·081		86	0·183	−0·076	
Paper	79	0·000	0·000	0·000	Machinery	79	0·000	0·000	
	80	−0·111	−0·035	−0·076		80	0·031	−0·025	
	81	−0·127	0·038	−0·165		81	0·125	0·070	
	82	−0·127	−0·079	−0·048		82	0·131	0·027	
	83	−0·084	−0·221	0·137		83	0·077	0·025	
	84	−0·073	−0·266	0·192		84	0·137	0·072	
	85	−0·252	−0·362	0·110		85	0·083	0·032	
	86	−0·131	−0·326	0·195		86	0·076	0·040	
All	79	0·000	0·000	0·000	Import competing	79	0·000	0·000	
	80	−0·010	0·018	−0·027		80	−0·063	0·027	[illegible]
	81	0·051	0·054	−0·003		81	0·032	0·092	[illegible]
	82	0·329	0·048	0·281		82	0·088	0·066	
	83	0·174	0·010	0·164		83	0·077	0·034	
	84	0·117	0·025	0·092		84	0·089	0·059	
	85	0·120	−0·003	0·123		85	0·095	0·061	
	86	0·193	0·066	0·127		86	0·319	0·107	
Export oriented	79	0·000	0·000	0·000	Nontraded	79	0·000	0·000	
	80	−0·059	−0·038	−0·021		80	0·044	0·021	
	81	−0·048	−0·054	0·006		81	0·101	0·047	
	82	0·591	0·040	0·551		82	0·228	0·038	
	83	0·326	0·015	0·311		83	0·127	−0·004	
	84	0·178	0·049	0·129		84	0·114	0·000	
	85	0·203	−0·011	0·214		85	0·101	−0·040	

Table 3: Decomposition of aggregate productivity growth.

6.5 Table 4: Plant-level productivity regressions

Read:

- Table 4 estimates the difference-in-differences-style productivity regressions in equation (12).
- The exit indicator is negative: exiting plants are about 8.1% less productive than surviving plants in the baseline specification.
- Import-competing-year interactions are positive and significant from 1981 onward.
- The import-competing productivity gains range from about 3% to 10.4% relative to nontraded sectors.
- Export-oriented sectors are more productive in levels, but they do not show the same persistent post-liberalization productivity improvement.
- Plant fixed-effect specifications deliver similar import-competing patterns, supporting the within-plant interpretation.

Economic message: This is the paper's central empirical table. It shows both selective exit and within-plant productivity improvements in import-competing sectors after trade liberalization.

TABLE 4

Estimates of equation 12

	(1)		(2)		(3)		(4)		(5)		(6)	
	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.
Export-oriented	0·106	0·030**	0·106	0·030**	0·112	0·031**	0·098	0·048**	0·095	0·048**	0·100	0·046**
Import-competing	0·105	0·021**	0·105	0·021**	0·103	0·021**	−0·024	0·040	−0·025	0·040	−0·007	0·039
ex_80	−0·054	0·025**	−0·053	0·025**	−0·055	0·025**	−0·071	0·026**	−0·068	0·026**	−0·071	0·026**
ex_81	−0·099	0·028**	−0·097	0·028**	−0·100	0·028**	−0·117	0·027**	−0·110	0·027**	−0·119	0·027**
ex_82	0·005	0·032	0·007	0·032	0·003	0·032	−0·054	0·028*	−0·042	0·028	−0·055	0·028*
ex_83	0·021	0·032	0·023	0·032	0·021	0·032	−0·036	0·029	−0·025	0·030	−0·038	0·029
ex_84	0·050	0·031	0·051	0·031	0·050	0·031	0·007	0·028	0·017	0·028	0·007	0·028
ex_85	0·030	0·030	0·032	0·031	0·028	0·030	−0·001	0·029	0·013	0·030	−0·003	0·029
ex_86					0·043	0·036					−0·008	0·034
im_80	0·011	0·014	0·011	0·014	0·010	0·014	0·013	0·014	0·013	0·014	0·013	0·014
im_81	0·047	0·015**	0·047	0·015**	0·046	0·015**	0·044	0·014**	0·044	0·014**	0·044	0·014**
im_82	0·033	0·016**	0·034	0·017**	0·030	0·016*	0·024	0·015*	0·024	0·015*	0·025	0·015*
im_83	0·042	0·017**	0·043	0·017**	0·043	0·017**	0·040	0·015**	0·041	0·015**	0·042	0·015**
im_84	0·062	0·017**	0·062	0·017**	0·063	0·017**	0·059	0·015**	0·059	0·015**	0·061	0·015**
im_85	0·103	0·017**	0·104	0·017**	0·104	0·017**	0·101	0·015**	0·102	0·016**	0·101	0·015**
im_86					0·071	0·019**					0·073	0·017**
Exit indicator	−0·081	0·011**	−0·076	0·014**			−0·019	0·010**	−0·010	0·013		
Exit_export indicator			−0·021	0·036					−0·069	0·035*		
Exit_import indicator			−0·007	0·023					−0·005	0·021		
Industry indicators	yes		yes		yes		yes		yes		yes	
Plant indicators	no		no		no		yes		yes		yes	
Year indicators	yes		yes		yes		yes		yes		yes	
R^2 (adjusted)	0·057		0·058		0·062		0·498		0·498		0·488	
N	22983		22983		25491		22983		22983		25491	

Note: ** and * indicate significance at a 5% and 10% level, respectively. Standard errors are corrected for heteroscedasticity. Standard errors in columns 1–3 are also adjusted for repeated observations on the same plant. Columns 1, 2, 4, and 5 do not include observations in 1986 because one cannot define exit for the last

Table 4: Estimates of equation (12).

6.6 Tables 5 and 6: Exchange-rate robustness checks

Read:

- Table 5 reports correlations between productivity growth and output growth. The correlations are small, ranging from 0.089 to 0.226 across industries.
- The small correlations weaken the view that measured productivity gains are mainly driven by demand expansions and spare capacity.
- Table 6 reports plant inventories by trade orientation over time.
- Inventory levels and inventory-output ratios do not fluctuate in a way that closely matches the timing of real-exchange-rate movements.

Economic message: The import-competing productivity gains are unlikely to be explained only by exchange-rate-induced demand movements or capacity utilization.

TABLE 5

Correlation between productivity growth and output growth

Industry	Correlation
Food processing	0·154
Textiles	0·226
Wood	0·089
Paper	0·156
Chemicals	0·150
Glass	0·106
Basic metals	0·109
Machinery	0·119

(a) Table 5: Productivity-output growth correlations

TABLE 6

Average plant inventories

	Inventories in levels			Inventories as a share of output		
Year	Export-oriented plants	Import-competing plants	Nontraded goods plants	Export-oriented plants	Import-competing plants	Nontraded goods plants
79	18,749	7,847	5,878	0·103	0·089	0·051
	(140,638)	(32,790)	(56,941)	(0·160)	(0·122)	(0·095)
80	21,210	6,252	5,266	0·093	0·085	0·052
	(143,646)	(19,889)	(50,525)	(0·151)	(0·133)	(0·108)
81	24,536	8,379	5,453	0·104	0·098	0·055
	(171,304)	(34,008)	(50,860)	(0·183)	(0·217)	(0·117)
82	39,259	7,716	6,053	0·109	0·117	0·053
	(253,373)	(23,937)	(57,764)	(0·228)	(0·196)	(0·123)
83	37,803	8,222	7,160	0·081	0·115	0·048
	(265,596)	(24,260)	(78,149)	(0·149)	(0·226)	(0·114)
84	44,123	10,536	7,931	0·079	0·114	0·145
	(228,173)	(34,205)	(81,188)	(0·141)	(0·252)	(0·101)
85	27,690	11,331	8,882	0·079	0·103	0·046
	(84,335)	(37,299)	(94,107)	(0·185)	(0·201)	(0·098)
86	24,530	11,612	7,626	0·062	0·077	0·038
	(77,253)	(37,453)	(61,588)	(0·120)	(0·110)	(0·080)
Overall	29,531	8,763	6,655	0·091	0·099	0·049
	(186,633)	(30,697)	(66,903)	(0·170)	(0·188)	(0·106)

Note: Inventories are measured as the end of the year plant-level inventories in

(b) Table 6: Average plant inventories

Tables 5 and 6: Robustness checks for demand and exchange-rate explanations.

6.7 Tables 7 and 8: Real exchange rate, tariffs, and import competition

Read:

- Table 7 relates productivity to the real exchange rate and its interactions with trade orientation.
- The real exchange rate is positively related to productivity in nontraded sectors and negatively related to export-oriented productivity, but the import interaction is insignificant.
- This suggests that exchange-rate movements are more relevant for the export-oriented sector than for the import-competing results.
- Table 8 shows that productivity is negatively related to tariffs and positively related to the import-output ratio.
- The import-output coefficient is positive and significant, reinforcing the conclusion that more import competition is associated with higher productivity.

Economic message: The results survive alternative ways of measuring exposure to trade. Lower protection and greater import competition are associated with higher measured productivity.

TABLE 7

Relationship between productivity and the real exchange rate

Real exchange rate	0·0006**
	(0·0001)
Real exchange rate * export indicator	−0·0011**
	(0·0003)
Real exchange rate * import indicator	0·0001
	(0·0001)
Plant indicators	Yes
R^2 (adjusted)	0·48

Note: ** and * indicate significance at a 5% and 10% level, respectively. Standard errors are corrected for heteroskedasticity. The regression also includes a time trend, export and import indicators, and the interac-

(a) Table 7: Productivity and real exchange rate

TABLE 8

Relationship between productivity and tariffs, real exchange rate, and import competition

	(1)	(2)	(3)	(4)
Real exchange rate		0·0005**		0·0005**
		(0·0001)		(0·0001)
Tariff	−0·2790**	−0·2377**		−0·2376**
	(0·0280)	(0·0286)		(0·0285)
Imports/output			0·0023**	0·0023**
			(0·0006)	(0·0006)
Plant indicators	yes	yes	yes	yes
R^2 (adjusted)	0·48	0·48	0·48	0·48

(b) Table 8: Productivity, tariffs, and imports/output

Tables 7 and 8: Robustness using exchange-rate and trade-exposure measures.

6.8 Appendix tables and Figure A.1: Data construction and trade orientation

Read:

- Table A1 reports descriptive statistics for output, value added, labour, capital, investment, and materials.
- Table A2 documents the panel structure: 3,470 plants in 1979, 2,508 plants in 1986, 25,491 plant-years, and 4,379 total plants.
- Table A3 reports trade-orientation measures at the three-digit ISIC level, including export-output ratios, import-output ratios, and import penetration.
- Figure A.1 plots export-output ratios against import-output ratios at the four-digit ISIC level. Most sectors are near one axis rather than high on both axes.
- This supports the paper's classification of industries into export-oriented, import-competing, and nontraded categories.

Economic message: The appendix clarifies that the empirical design depends on careful construction of plant inputs, capital, and trade-orientation categories. The trade-orientation classification is not arbitrary; most industries have a clearly dominant import or export orientation.

979 to 1986 on a four-digit ISIC level, and 0·35 on a three-digi
'as 0·257, the median export–output ratio was 0·017. As Table A.:

TABLE A1

Descriptive statistics

Variable	Mean	S.D.	Median
Output	132,165	985,792	17,922
Value added	50,386	325,322	6,276
Labour	56	121	23
Skilled labour	15	41	5
Unskilled labour	41	87	18
Capital	53,066	360,274	3,682
Investment	3,861	41,624	0
Materials	73,337	700,712	9,643

Note: Quantities in thousands of 1980 pesos. Labour is

(a) Table A1: Descriptive statistics

TABLE A2

Panel information

Year	Number of plants	Years in the panel	Number of plants
1979	3,470	8	1,536
1980	3,704	7	716
1981	3,654	6	406
1982	3,396	5	400
1983	3,034	4	441
1984	2,928	3	385
1985	2,797	2	341
1986	2,508	1	154
Total number of plant-years	25,491	Total number of plants	4,379

Note: The left hand side of the table indicates the number of plants in a given year. The right hand side of the table indicates the number of plants that stay in the panel for a total of 8,7,... years.

(b) Table A2: Panel information

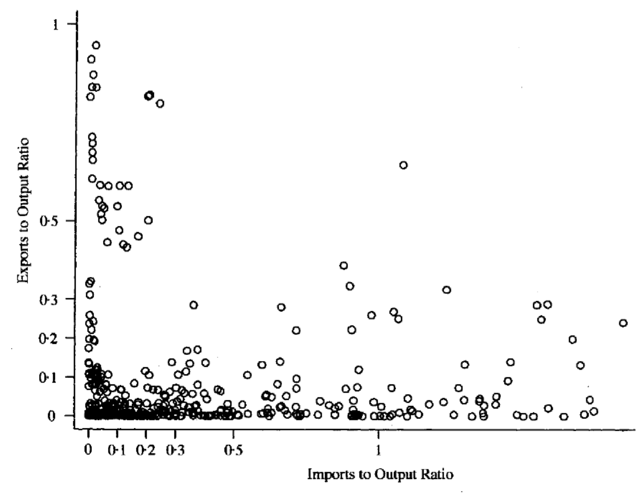
Appendix Tables A1 and A2.

TABLE A3

3-digit ISIC industry trade orientation

Industry	Export–output ratio	Import–output ratio	Imp Peneti
312	0·174	0·078	0·0
313	0·046	0·045	0·0
321	0·006	0·271	0·2
322	0·004	0·174	0·1
323	0·008	0·135	0·1
324	0·004	0·097	0·0
331	0·254	0·019	0·0
332	0·016	0·089	0·0
341	0·418	0·096	0·0
342	0·023	0·062	0·0
351	0·824	1·326	0·5
352	0·002	0·113	0·1
353	0·029	0·114	0·0
354	0·019	0·262	0·1
355	0·040	0·296	0·2
356	0·002	0·102	0·0
361	0·092	0·716	0·3
362	0·017	0·318	0·2
369	0·003	0·078	0·0
371	0·063	0·112	0·1
372	0·733	0·012	0·0
381	0·097	0·255	0·2
382	0·053	2·141	0·6
383	0·036	1·649	0·6
384	0·210	2·010	0·6
385	0·089	7·381	0·8

(a) Table A3: ISIC trade orientation



Source: Statistics Canada

FIGURE A.1

(b) Figure A.1: Export-output vs. import-output ratios

Appendix trade-orientation evidence.

7 Short summary

- The paper studies Chilean manufacturing plants from 1979 to 1986 after a major trade liberalization.
- It asks whether liberalized trade improves productivity and through which channels.
- Productivity is estimated using a semiparametric production-function method that corrects for simultaneity and selection from plant exit.
- The key treatment comparison is import-competing versus nontraded sectors over time.
- Import-competing plants become about 3% to 10% more productive relative to nontraded plants.
- Exiting plants are about 8% less productive than surviving plants.
- Aggregate productivity growth is driven substantially by reallocating output toward more productive plants.
- Robustness checks using tariffs, import-output ratios, and exchange rates support the import-competition interpretation.
- The main contribution is to show that trade liberalization affects productivity through within-plant discipline, selective exit, and reallocation, not just through representative-firm scale effects.

8 Conclusion

8.1 Core conclusion

Pavcnik shows that trade liberalization in Chile is associated with significant productivity improvements among plants in import-competing sectors. The main plant-level result is that these plants became 3% to 10% more productive than plants in nontraded sectors. The main aggregate result is that productivity growth also reflects exit and reallocation: less productive plants leave, and more productive plants account for a rising share of output.

8.2 Economic intuition

Before trade liberalization, high protection allowed heterogeneous plants to coexist, including relatively inefficient producers. When protection fell, import competition lowered the tolerance for inefficiency. Some plants survived by improving productivity, possibly by reducing X-inefficiency, reorganizing production, adopting better inputs, or facing stronger competitive discipline. Other plants exited. At the same time, resources and market shares shifted toward more productive plants. These three mechanisms together explain why aggregate productivity rose.

8.3 Incremental contribution revisited

The paper's incremental contribution is both substantive and methodological. Substantively, it documents that trade liberalization can improve productivity through plant heterogeneity, exit, and reallocation. Methodologically, it demonstrates that credible productivity measurement requires correcting for input endogeneity and selection from exit. Empirically, it improves identification by comparing traded and nontraded sectors rather than relying only on before-after reform comparisons.

8.4 Implications for trade and industrial policy

The findings imply that the gains from trade liberalization depend on the economy's ability to reallocate resources. If inefficient plants cannot exit, or if labour and capital cannot move toward more productive producers, trade liberalization may deliver smaller productivity gains. Conversely, institutions that allow plant turnover, competition, and resource mobility can amplify the benefits of openness.

8.5 Limitations

The paper does not observe plant-level physical output prices, so measured productivity may include markup variation. The sample begins after the main tariff reductions, so pre-liberalization plant-level trends are not observed. The trade-orientation classification is useful but imperfect, and sector-specific shocks could still contaminate the difference-in-differences estimates. Finally, the paper identifies relative productivity improvements associated with trade exposure, not the full welfare effect of liberalization including displaced workers and adjustment costs.

8.6 Takeaway

The enduring takeaway is that trade liberalization works through selection and reallocation in addition to within-plant productivity improvements. The paper is an early empirical bridge between trade liberalization and the heterogeneous-firm view of productivity: opening to trade disciplines protected producers, forces inefficient plants out, and reallocates economic activity toward more productive plants.